The Gender Gap in Confidence: Expected but Not Accounted For

Heterogeneity in Evaluator Beliefs

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Abstract

I replicate the main result from Exley and Nielsen (2024), who show that the gender gap in confidence is "contagious"—causing evaluators to hold overly pessimistic posterior beliefs about women's performance. Their baseline estimate of the expected performance gap is 10.49 percentage points, roughly six times the true gap of 1.74 percentage points. I then conduct an internal meta-analysis to ask whether this overestimation is uniform across evaluator types or whether certain demographic and cognitive characteristics predict a larger gap. Across 14 subgroup regressions, I find that the overestimation is pervasive: every subgroup produces an estimate far above the true gap. A weighted meta-regression with $R^2 = 0.876$ reveals that evaluators who exhibit base rate neglect, are older, lean Republican, or identify as female produce systematically larger gaps. The results suggest that the contagious confidence gap is not driven by a single type of evaluator, but that susceptibility to base rate neglect is the single strongest predictor of the bias.

1 Introduction

One of the most robust findings in the behavioral economics literature is that women, on average, express less confidence in their own performance than equally performing men do—a pattern documented across a wide range of settings and tasks (Lundeberg et al. 1994; Mobius et al. 2022). What is less well understood is how this confidence gap affects the people who observe and evaluate women's self-assessments. If evaluators—employers, colleagues, supervisors—correctly anticipate and adjust for the fact that women tend to understate their abilities, the confidence gap need not translate into worse outcomes for women. But if evaluators fail to make that adjustment, the confidence gap becomes, in the language of Exley and Nielsen (2024), "contagious."

Exley and Nielsen (2024) provide clean experimental evidence that the confidence gap is indeed contagious. In their design, evaluators are shown gender-disaggregated self-evaluation data and then asked to report their posterior beliefs about a randomly selected worker's performance. Even though evaluators demonstrably expect the confidence gap—they know women tend to understate their performance—they still update too strongly in the direction of the self-evaluation information. The result is a posterior expected performance gap of roughly 10.5 percentage points, more than six times the true gap of 1.74 percentage points. The authors trace this failure to a "calculation problem": evaluators are unable or unwilling to perform the Bayesian updating required to correctly discount the self-evaluation signal.

This paper takes that finding as a starting point and asks a natural follow-up question: *is the contagious confidence gap equally strong across all types of evaluators, or does it vary systematically with evaluator characteristics?* Understanding heterogeneity in this bias matters for at least two reasons. First, if the bias is concentrated among particular groups—say, evaluators who score low on cognitive reflection tests or who exhibit base rate neglect—then targeted interventions might be more cost-effective than broad ones. Second, the pattern of heterogeneity itself can shed light on the mechanism: if the gap is larger

precisely among evaluators who are prone to base rate neglect, that is additional evidence that the calculation problem, rather than taste-based discrimination, is the operative channel.

To answer this question, I use the internal meta-analysis framework described in Banzhaf and Smith (2007) and introduced in the course tutorial. I re-estimate the authors' main regression specification on 14 subgroups defined by evaluator gender, age, cognitive reflection test (CRT) score, base rate neglect, education, income, and political affiliation. I then treat the 14 resulting point estimates as outcomes in a weighted meta-regression, where the weights are inverse-variance weights and the regressors are indicators for each subgroup characteristic.

The main findings are as follows. First, the contagious confidence gap is universal: every single subgroup produces an estimated expected performance gap well above the true gap of 1.74 percentage points, with all 14 estimates statistically significant at the 1 percent level. Second, there is meaningful variation around the full-sample estimate of 10.49 percentage points, ranging from 9.19 pp (young evaluators) to 12.66 pp (evaluators who exhibit base rate neglect). Third, the meta-regression explains 87.6 percent of this variation with seven evaluator-level indicators. Base rate neglect is the strongest predictor, adding roughly 2.81 percentage points to the expected gap; being older adds 2.19 pp; and leaning Republican adds 1.96 pp. The CRT score indicator is the only one that falls short of conventional significance thresholds.

The rest of the paper is organized as follows. Section 2 describes the data. Section 3 presents the econometric model. Section 4 describes the meta-analysis design. Section 5 presents the results. Section 6 concludes.

2 Data

The data come from the Evaluator Study in Exley and Nielsen (2024), which is publicly available in the authors' replication repository. The study was conducted on Prolific, an

online survey platform, and recruited 2,400 evaluators who were randomly assigned to one of six treatments. For the purposes of this paper, I restrict attention to the three "Gender-Known" treatments—Baseline, Attention, and Calculation—which together comprise 1,210 observations. These are the treatments in which evaluators are explicitly told whether their assigned worker is male or female, making the gender gap in posterior beliefs directly interpretable.

The key outcome variable is `post`, the evaluator's posterior belief about the percent chance that their assigned worker has a "poor performance" on the math and science test. This is elicited after the evaluator is shown accurate aggregate self-evaluation data: evaluators assigned a female worker are told that 80 percent of female workers predicted a poor performance, while evaluators assigned a male worker are told that 56 percent of male workers did so. The key regressor of interest is `bfemale`, an indicator equal to one if the evaluator was assigned a female worker.

The true performance gap—the actual difference in the probability of a poor performance between female and male workers—is 1.74 percentage points. This serves as the benchmark throughout the analysis. In the analysis sample, evaluators assigned a female worker report a mean posterior belief of 56.29 percentage points, compared to 47.50 percentage points for evaluators assigned a male worker, yielding a raw gap of 8.79 pp that is consistent with the authors' regression-adjusted estimate of 10.49 pp.

For the subgroup analysis, I use the following evaluator-level characteristics available in the data:

- **Evaluator gender** (`female`, `male`): whether the evaluator self-identifies as female or male. Note that 34 evaluators identify as neither male nor female and are excluded from the gender subgroup regressions but remain in all other subgroup analyses; the gender subgroups thus sum to 1,176.
- **Age** (`young`, `old`): split at the sample median of 33 years.

- **Cognitive Reflection Test score** (`high_crt`, `low_crt`): `high_crt` is defined as a CRT score of 2 or above (out of 3); `low_crt` is a score of 0 or 1.
- **Base rate neglect** (`base_pn`): an indicator equal to one if the evaluator exhibited pure base rate neglect on the incentivized bonus question at the end of the study.
- **Education** (`educLow`, `educHigh`): split at the sample median.
- **Income** (`incomeLow`, `incomeHigh`): split at the sample median.
- **Political affiliation** (`favor_dem`, `wfavor_rep`): whether the evaluator leans Democratic or Republican.

Table 1 provides summary statistics for the full analysis sample. The sample is roughly gender-balanced among binary-identifying evaluators (669 female, 507 male, 34 other), with 55.3 percent identifying as female.

Table 1: Summary Statistics: Evaluator Study, Gender-Known Treatments

Variable	Mean	Std. Dev.	Min	Max
Posterior belief (`post`)	51.92	18.96	0	100
Female worker indicator (`bfemale`)	0.502	0.500	0	1
Female evaluator (`female`)	0.553	0.497	0	1
High CRT score ($\geq$ 2 out of 3)	0.547	0.498	0	1
Base rate neglect (`base_pn`)	0.296	0.457	0	1
Democrat-leaning (`favor_dem`)	0.683	0.466	0	1
Republican-leaning (`wfavor_rep`)	0.317	0.466	0	1
Observations		1,210		

Notes: Sample restricted to Gender-Known treatments (Baseline, Attention, Calculation). CRT score is from the three-question Cognitive Reflection Test; `high_crt` indicates a score of 2 or 3. Base rate neglect indicator equals one if the evaluator's answer to the incentivized base rate question was consistent with ignoring the prior entirely. The female evaluator mean is computed over all 1,210 observations; 34 evaluators identify as neither male nor female.

3 Model Specification

The focal metric in this paper is the coefficient on `bfemale` from the authors' main regression, which I refer to as $\hat{\delta}$. The regression is:

$$\text{post}_i = \alpha + \delta \cdot \text{bfemale}_i + \beta_1 \cdot \text{t_attn}_i + \beta_2 \cdot \text{t_calc}_i + \gamma_1 \cdot (\text{t_attn} \times \text{bfemale})_i + \gamma_2 \cdot (\text{t_calc} \times \text{bfemale})_i + \varepsilon_i \tag{1}$$

where post_i is evaluator i's posterior belief (in percentage points) about the probability that their assigned worker has a poor performance; bfemale_i is an indicator equal to one if evaluator i was assigned a female worker; t_attn_i and t_calc_i are indicators for the Attention and Calculation treatments, respectively; and the interaction terms allow the treatment effects to differ by worker gender. Standard errors are heteroskedasticity-robust throughout.

The coefficient of interest is δ, which captures the expected performance gap in the Baseline treatment—that is, how much more likely evaluators believe a female worker is to have a poor performance relative to a male worker, after seeing the self-evaluation data. Under correct Bayesian updating, this gap should be small and statistically indistinguishable from the true gap of 1.74 percentage points. Exley and Nielsen (2024) find $\hat{\delta} = 10.49$ pp (SE = 1.78), which is more than six times the true gap and highly statistically significant.

I replicate this estimate exactly. The replication serves as the anchor for the internal meta-analysis: I re-estimate equation (1) on 14 subsamples defined by evaluator characteristics, and I treat each resulting $\hat{\delta}_s$ as an outcome in the meta-regression described in Section 4.

4 Analysis Design

The goal of the internal meta-analysis is to understand how the expected performance gap $\hat{\delta}$ varies across evaluator subgroups. Following the framework in Banzhaf and Smith (2007), I

treat each subgroup estimate as a data point and ask whether observable characteristics of the subgroup predict the magnitude of the gap.

4.1 Subgroup Regressions

I estimate equation (1) separately for 14 subgroups defined by seven evaluator characteristics: gender (female, male), age (young, old), CRT score (high, low), base rate neglect (yes, no), education (low, high), income (low, high), and political affiliation (Democrat, Republican). Each subgroup regression produces a point estimate $\hat{\delta}_s$ and a standard error $\widehat{SE}_s$.

The choice of subgroups is motivated by the authors' own heterogeneity analysis and by the theoretical mechanism they propose. If the contagious confidence gap is driven primarily by failures in Bayesian updating, then evaluators who are more prone to cognitive biases—particularly base rate neglect—should exhibit a larger gap. Similarly, if the gap reflects any residual role for taste-based discrimination or gender stereotyping, we might expect it to be larger among male evaluators or among evaluators with more traditional social views. The age, education, and income splits are included as additional demographic checks.

4.2 Meta-Regression

I then estimate a weighted meta-regression of the form:

$$\hat{\delta}_s = \mu + \sum_{k=1}^{K} \lambda_k \cdot D_{ks} + u_s, \qquad w_s = \frac{1}{\widehat{SE}_s^2} \tag{2}$$

where $\hat{\delta}_s$ is the estimated expected performance gap in subgroup s; D_{ks} is an indicator equal to one if subgroup s belongs to the "lower" or "different" category for characteristic k (e.g., $D_{ks} = 1$ for female evaluators, old evaluators, low-CRT evaluators, etc.); μ is the intercept, interpreted as the expected gap for the reference group (male, young, high-CRT, no base rate neglect, high education, high income, Democrat-leaning); and $w_s = 1/\widehat{SE}_s^2$ is

the inverse-variance weight, which gives more weight to more precisely estimated subgroup effects.

The reference category for each characteristic is chosen to be the group that, a priori, might be expected to exhibit the smallest gap: male evaluators (less likely to have personal experience with the confidence gap), young evaluators, high-CRT evaluators (better at Bayesian reasoning), evaluators without base rate neglect, high-education evaluators, high-income evaluators, and Democrat-leaning evaluators.

The meta-regression has 15 observations (14 subgroups plus the full sample) and 8 parameters (intercept plus 7 indicators), leaving 7 degrees of freedom. While the sample is small, the R^2 of 0.876 suggests that the model fits well and that the chosen characteristics capture most of the systematic variation in $\hat{\delta}_s$ across subgroups.

5 Analysis Results

5.1 Subgroup Estimates

Table 2 reports the estimated expected performance gap $\hat{\delta}_s$ for each of the 14 subgroups, along with the corresponding standard error, sample size, and 95 percent confidence interval. Several patterns are immediately apparent.

First, and most strikingly, the overestimation is universal. Every single subgroup produces an estimate well above the true gap of 1.74 pp, and in every case the estimate is statistically significant at the 1 percent level. The smallest estimate is 9.19 pp (young evaluators), which is still more than five times the true gap. This suggests that the contagious confidence gap is not a phenomenon driven by any particular type of evaluator—it is pervasive.

Second, there is meaningful variation across subgroups. The range of estimates spans from 9.19 pp to 12.66 pp, a difference of about 3.5 percentage points. While this variation is modest relative to the overall level of the bias, it is systematic, as the meta-regression results

below show.

Third, the subgroups that produce the largest gaps are those that the calculation-problem mechanism would predict: evaluators who exhibit base rate neglect (12.66 pp), older evaluators (12.04 pp), and Republican-leaning evaluators (11.81 pp). The subgroups that produce the smallest gaps—young evaluators (9.19 pp), male evaluators (9.31 pp), and evaluators without base rate neglect (9.51 pp)—are also consistent with this pattern.

Table 2: Subgroup Estimates of the Expected Performance Gap

Subgroup	$\hat{\delta}_s$	SE	N	95% CI
Panel A: Evaluator Gender				
Female evaluators	11.56	2.27	669	[7.11, 16.00]
Male evaluators	9.31	3.02	507	[3.38, 15.24]
Panel B: Age				
Young evaluators	9.19	2.32	691	[4.64, 13.74]
Old evaluators	12.04	2.74	519	[6.66, 17.42]
Panel C: Cognitive Reflection Test				
High CRT	10.10	2.36	662	[5.47, 14.72]
Low CRT	10.82	2.71	548	[5.51, 16.13]
Panel D: Base Rate Neglect				
No base rate neglect	9.51	2.14	852	[5.31, 13.71]
Base rate neglect	12.66	3.19	358	[6.41, 18.91]
Panel E: Education				
High education	9.90	2.45	638	[5.10, 14.71]
Low education	11.20	2.61	572	[6.09, 16.31]
Panel F: Income				
High income	9.94	2.20	679	[5.63, 14.25]
Low income	11.33	3.00	531	[5.45, 17.21]
Panel G: Political Affiliation				
Democrat-leaning	9.78	2.15	826	[5.57, 14.00]
Republican-leaning	11.81	3.17	384	[5.60, 18.02]
Full sample	10.49	1.78	1,210	[7.00, 13.98]
True gap	1.74	—	—	—

Notes: Each row reports the OLS estimate of $\hat{\delta}_s$ from equation (1) estimated on the indicated subgroup. All regressions include treatment indicators and their interactions with `bfemale`. Standard errors are heteroskedasticity-robust. Gender subgroups sum to 1,176 because 34 evaluators identify as neither male nor female. The true performance gap is 1.74 pp, as reported in Exley and Nielsen (2024).

5.2 Forest Plot

Figure 1 displays the subgroup estimates and their 95 percent confidence intervals in a forest plot. The red dashed vertical line marks the true performance gap of 1.74 pp. The

figure makes visually clear that every confidence interval lies entirely to the right of the true gap, confirming that the overestimation is statistically significant in every subgroup. The clustering of estimates between 9 and 13 pp, all far from the true gap, reinforces the conclusion that the contagious confidence gap is a robust, pervasive phenomenon rather than an artifact of any particular subgroup.

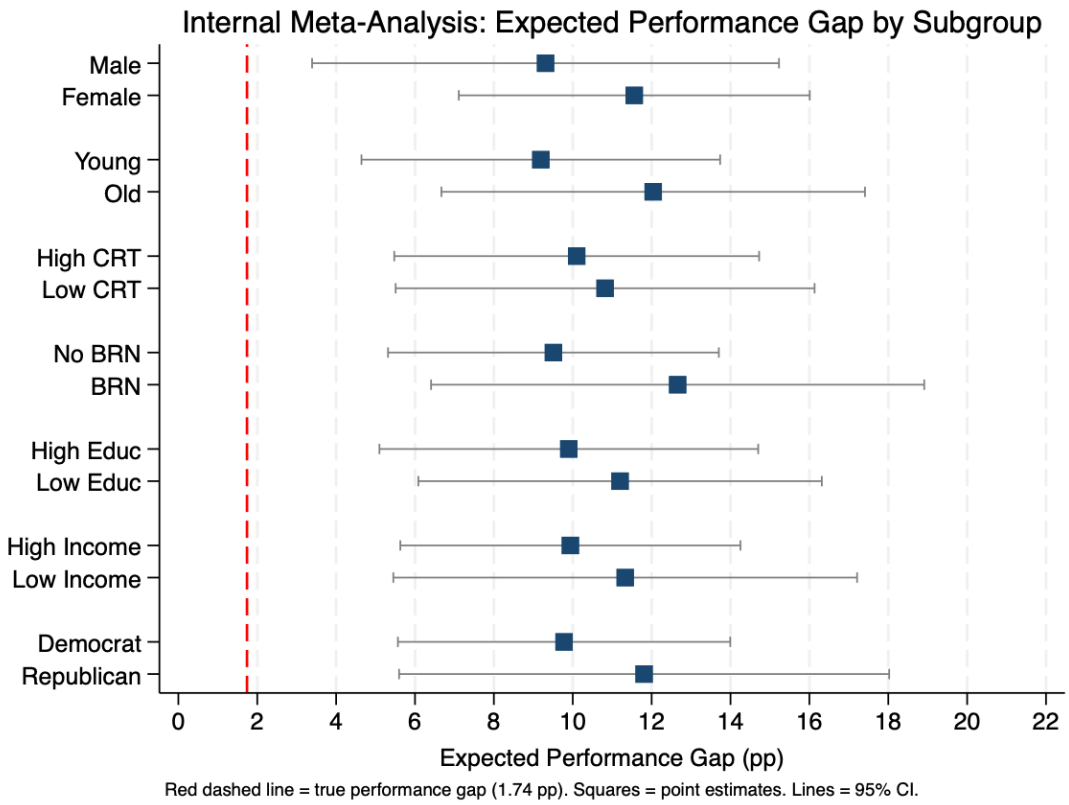

Figure 1: Internal Meta-Analysis: Expected Performance Gap by Evaluator Subgroup

Notes: Each square represents the OLS point estimate of $\hat{\delta}_s$ from equation (1) estimated on the indicated subgroup. Horizontal lines are 95 percent confidence intervals. The red dashed vertical line marks the true performance gap of 1.74 pp. Subgroups are grouped by characteristic: evaluator gender (top), age, CRT score, base rate neglect, education, income, and political affiliation (bottom).

5.3 Meta-Regression

To formally test which evaluator characteristics predict the magnitude of the contagious confidence gap, I estimate the weighted meta-regression introduced in equation (2), reproduced

here for convenience:

$$\hat{\delta}_s = \mu + \sum_{k=1}^{K} \lambda_k \cdot D_{ks} + u_s, \qquad w_s = \frac{1}{\widehat{SE}_s^2}$$

Table 3 reports the results. The model fits well, with $R^2 = 0.876$ and an F-statistic of 7.09 ($p = 0.010$), suggesting that the seven evaluator-level indicators jointly explain most of the variation in $\hat{\delta}_s$ across subgroups.

The intercept of 9.85 pp represents the estimated expected performance gap for the reference group (male, young, high-CRT, no base rate neglect, high education, high income, Democrat-leaning). This is itself far above the true gap of 1.74 pp, which again underscores that the bias is not confined to any particular group.

Among the seven indicators, six are statistically significant at conventional levels. The largest coefficient belongs to base rate neglect: evaluators who exhibit pure base rate neglect produce a gap that is 2.81 pp larger than those who do not ($p = 0.004$). This is the most direct evidence linking the contagious confidence gap to a specific cognitive failure. Being older adds 2.19 pp ($p = 0.007$), leaning Republican adds 1.96 pp ($p = 0.020$), being female adds 1.71 pp ($p = 0.009$), having lower income adds 1.48 pp ($p = 0.049$), and having lower education adds 1.35 pp ($p = 0.043$). The only indicator that falls short of significance is low CRT score, which adds 0.97 pp ($p = 0.131$).

The finding that female evaluators produce a *larger* gap than male evaluators is consistent with the authors' own heterogeneity result (their Table 6), which shows that female evaluators are at least as likely as male evaluators to hold posterior beliefs that disfavor women. This pattern cuts against a taste-based discrimination explanation and is more consistent with the calculation-problem story: female evaluators are not immune to the bias, and may in fact be slightly more susceptible to it.

The insignificance of the CRT coefficient is perhaps the most surprising finding. One might

expect that evaluators who score higher on the CRT—a measure of reflective, deliberate thinking—would be better at Bayesian updating and therefore exhibit a smaller gap. The near-zero and insignificant coefficient suggests that general cognitive ability, as measured by the CRT, does not translate into better performance on this specific updating task. This is consistent with the authors' own Table 5, which shows that the expected performance gap is if anything *directionally larger* for high-CRT evaluators. The calculation problem, it seems, is hard enough that even reflective thinkers struggle with it.

Table 3: Weighted Meta-Regression: Predictors of the Expected Performance Gap

	Coefficient	Std. Error
Female evaluator (D_{female})	1.708***	(0.482)
Old evaluator (D_{old})	2.188***	(0.572)
Low CRT ($D_{\text{low_crt}}$)	0.968	(0.566)
Base rate neglect (D_{BRN})	2.808***	(0.659)
Low education ($D_{\text{low_educ}}$)	1.348**	(0.547)
Low income ($D_{\text{low_income}}$)	1.478**	(0.622)
Republican-leaning (D_{rep})	1.958**	(0.656)
Constant (μ)	9.852***	(0.159)
Observations	15	
R^2	0.876	
Adj. R^2	0.753	
F-statistic	7.09	

Notes: Weighted OLS with weights $w_s = 1/\widehat{SE}_s^2$. The dependent variable is $\hat{\delta}_s$, the estimated expected performance gap from equation (1) for subgroup s. Each indicator equals one for the named group and zero for its complement. The reference group is male, young, high-CRT, no base rate neglect, high education, high income, Democrat-leaning. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6 Conclusions

This paper replicates the central finding of Exley and Nielsen (2024)—that the gender gap in confidence is contagious, causing evaluators to hold posterior beliefs about women's performance that are far more pessimistic than the truth—and then asks whether this bias is

uniform or heterogeneous across evaluator types. The answer is both. The bias is universal: no subgroup of evaluators comes close to the true performance gap of 1.74 pp, and every subgroup estimate is statistically significant. But the bias is also heterogeneous in predictable ways: evaluators who exhibit base rate neglect, who are older, who lean Republican, or who identify as female all produce systematically larger gaps.

The most economically meaningful finding is the role of base rate neglect. The fact that evaluators who demonstrably ignore prior probabilities in an unrelated task also exhibit a larger contagious confidence gap provides additional, independent evidence that the mechanism is a calculation problem—a failure of Bayesian updating—rather than taste-based discrimination. This has direct implications for intervention design. The Calculation treatment in Exley and Nielsen (2024), which provides evaluators with their implied Bayesian posteriors before eliciting their beliefs, is the only intervention that significantly reduces the gap. The present results suggest that this type of intervention might be especially valuable when targeted at evaluators who are prone to base rate neglect.

Several limitations are worth noting. The subgroup sample sizes are relatively small, particularly for the base rate neglect and Republican subgroups, which reduces precision. The meta-regression itself has only 15 observations, so the coefficient estimates should be interpreted with appropriate caution. Future work could expand the set of subgroups, explore interactions between characteristics (e.g., whether base rate neglect matters more for older evaluators), and test whether the heterogeneity patterns documented here replicate in field settings with professional evaluators.

References

Banzhaf, H. Spencer and V. Kerry Smith, "Meta-Analysis in Model Implementation: Choice Sets and the Valuation of Air Quality Improvements," *Journal of Applied Econometrics*, 2007, *22* (6), 1013–1031.

Exley, Christine L. and Kirby Nielsen, "The Gender Gap in Confidence: Expected but Not Accounted For," *American Economic Review*, 2024, *114* (3), 851–885.

Lundeberg, Mary A., Paul W. Fox, and Judith Punčcohař, "Highly Confident but Wrong: Gender Differences and Similarities in Confidence Judgments," *Journal of Educational Psychology*, 1994, *86* (1), 114–121.

Mobius, Markus, Muriel Niederle, Paul Niehaus, and Tanya Rosenblat, "Managing Self-Confidence: Theory and Experimental Evidence," *Management Science*, 2022, *68* (11), 7793–7817.